MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NUMERICAL INDICATOR

TYPES 8037 AND 8421

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Gas filled, cold cathode containing a mercury capsule. The 8421 affords better visibility (wide angle view) of the numerals.

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ebb	Ik (Individual)	Rp	TA	Alt
Unit:	Vdc	mA	Ohms	C	ft
Maximum:	---	3.5	---	+55	70,000
Minimum:	170	---	---	-20	---
TEST CONDITIONS:	170	---	10 K ± 1%	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
1006	Salt spray (corrosion)		---	---	---	---
	Quality conformance inspection, part 1					
3347	Ionization voltage (1) (with illumination)	Ambient illumination = 5 to 50 foot candles	Ebb	---	170	Vdc
3333	Uniformity of cathode glow	Ambient illumination = 5 to 50 foot candles (see note 1)	---	---	---	---
1256	Electrode current (cathode)	Individual cathodes (see note 1)	Ik	1.5	3.0	mA
	Quality conformance inspection, part 2					
1002	Barometric pressure, reduced	eb = 170; Rp = 10 K; pressure = 35 mmHg (see notes 2 and 3)	---	---	---	---
1031	Variable-frequency vibration	See notes 4 and 5	---	---	---	---
---	Light output	Light individual cathode (see notes 1 and 6)	---	0.5	---	ftC
---	Alinement, digits	See note 7	---	---	±3	Degrees
---	Shock	200 G; duration = 1.5 ±.1 ms (see notes 4 and 6)	---	---	---	---
1105	Permanence of marking		---	---	---	---
	Quality conformance inspection, part 3					
---	Life test (1)	Group C; sequential counting; 1 Hz; (see note 9); t = 1,000 hours	---	---	---	---
---	Life-test (1) end points	Uniformity of cathode glow (see note 1)	---	---	---	---
		Cathode current	Ik	1.3	3.3	mA
		Light output	---	0.5	---	ftC
---	Life test (2)	Group C; one numeral cathode glowing steadily (see notes 10, 11, and 12), t = 1,000 hours minimum	---	---	---	---
---	Life-test (2) end points	Uniformity of cathode glow, numeral under test (see note 1)	---	---	---	---
		Cathode current, numeral under test	Ik	1.3	3.6	mA
		Light output, numeral under test	---	0.5	---	ftC

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NOTES:

1. No glow shall be permitted on any tube part other than the numeral under test.

2. The tube shall be tested by using a 60-cycle sine wave, full-wave rectified and unfiltered, having the indicated peak value.

3. Pins No. 1 and 8 shall not be used for this test.

4. No permanent shorts or opens shall be permitted.

5. On each plane, the 10 cathodes shall be sequentially energized at the test conditions.

6. Light output of each numeral is measured with a foot-candle (ftC) meter and an eye-corrected photronic cell. The entire cell area is illuminated and is coupled to the tube by a light-tight cylinder so that the face of the cell is 1-1/2 inches (38.10 mm) from the tube base.

7. Centerline of numerals, drawn through the top of the numeral support posts, shall be within the tolerance specified herein in relation to line drawn on the base of the tube through pins 1 and 8.

8. Shock test shall be made on any shock machine capable of producing a half-sine wave shock waveform of the specified duration and amplitude. No voltages shall be applied during this test. Each tube shall be subjected to a total of 20 shocks, i.e., five shocks in each of positions X, Y, Z plus, and Z minus in any sequence. This is a destructive test. Tubes subjected to destructive tests shall not be delivered on the contract or order.

9. Tubes shall be energized with a duty cycle greater than 90 percent.

10. Group C, life-test sampling plan, and Group C, establishment of prerelease eligibility (MIL-E-1 paragraphs) should be used with the following exceptions:

    (a) Group C life-test sampling plan.

        (1) The first sample size = 3.
        (2) The first cumulative sample size = 3.
        (3) The second sample size = 3.
        (4) The second cumulative sample size = 6.

    (b) Group C, establishment of prerelease eligibility.

        (1) A minimum of 15 tubes shall-------etc.
        (2) Replace all reference to the current 4-tube sample with current 3-tube sample. Replace all reference to the last 16 with the last 15.

11. One tube shall have the top numeral cathode glowing steadily. One tube shall have the bottom numeral cathode glowing steadily. One tube shall have the "0" numeral cathode glowing steadily.

12. The life-test (2) sample from the first lot each month shall continue on life test for an additional 1,000 hours (2,000-hours total life-test time). Two thousand hour life-test end points shall be the same as the end points for 1,000-hour life test (2). Failure of this 2,000-hour life test shall result in the following action: Acceptance of subsequent lots shall be dependent on each lot passing the 2,000-hour life test until such time as 3 consecutive lots pass the 2,000-hour life test. At this time acceptance will revert to the 1,000-hour life test with only the first lot each month continuing to 2,000 as before.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - CE, SM
Air Force - 11, 17
DSA - ES

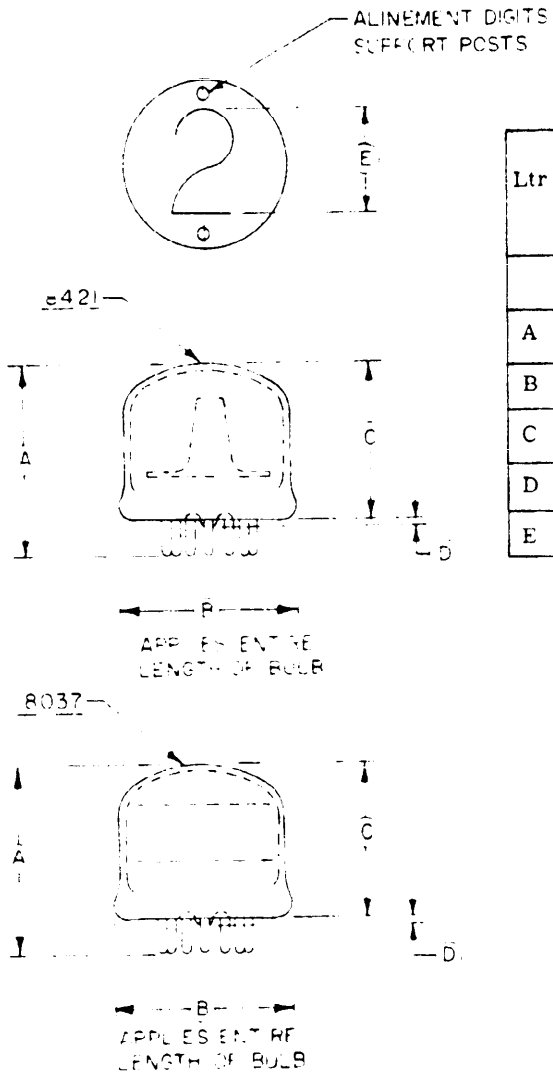
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2533)

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	1.250 (31.75)	1.380 (35.05)
B		1.080 (27.43)
C	1.000 (25.40)	1.124 (28.55)
D		.064 (1.63)
E	.595 (15.11)	.625 (15.88)

PIN NO.	ELEMENT
1	INT CON
2	ANODE
3	k0
4	k9
5	k8
6	k7
7	k6
8	INT CON
9	k5
10	k4
11	k3
12	k2
13	k1



NOTE:
a. Dashed lines shown on tube outlines are provided for information only.

FIGURE 1. Outline drawing of electron tube types 8037 and 8421.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1454C ELECTRON TUBE, NUMERICAL INDICATOR TYPES 8037 AND 8421

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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